

# Gregory Harutyunyan

AI / Machine Learning Engineer

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## Summary

**Mathematics Graduate** with an **MSc in AI + Data Engineering** in its final stage. Hands-on experience developing a production-oriented LLM inference backend for HPC deployment (Qualígrafo). Applied expertise in machine learning, NLP, RAG/KAG systems, computer vision, deep learning (CNNs, graph neural networks), anomaly detection and large-scale pipelines (Spark, Kafka, BigQuery). **Available for internships until July 2027.**

## Experience

### Backend AI Engineer / LLMOps - Internship

Mar 2026 – May 2026

Qualígrafo S.L. · Mimesis Platform

3 months · Granada (Remote)

- **AI Assistant Backend (HPC):** Migrated the API of an AI assistant for psychologists and researchers (2 mixed-methods pipelines of 4 phases each, with 10 endpoints) to an asynchronous job-based system. Validated the architecture for the SCAYLE supercomputing cluster.
- **Hallucination mitigation:** Implemented JSON Schemas and real-time validation with Pydantic for 8 prompt stages in an orchestrated LLM pipeline.
- **Inference flexibility:** Developed an abstraction layer with LangChain to switch between 3 LLM providers (Ollama, Groq, Azure OpenAI), with ~2s inference latency.

## Skills

**Languages & Backend:** Python, SQL, FastAPI, Pydantic, MATLAB.

**ML, NLP & Vision:** Scikit-learn, LightGBM, Transformers (Hugging Face), spaCy, LLMs (Llama 3.3, Ollama, Groq), LangChain, RAG/KAG, Pandas, NumPy, Computer Vision, Unsupervised Learning (PCA, Isolation Forest).

**Deep Learning:** TensorFlow/Keras (MLPs, CNNs), PyTorch, GNN (GraphSAGE, GIN, GATv2).

**Quantum Computing:** Qiskit, Qiskit Aer, IBM Quantum Runtime.

**Cloud & Data Engineering:** GCP (BigQuery, Cloud Storage, Dataproc), Apache Spark/Kafka.

**Infrastructure & MLOps:** Git, Linux, Docker, MLflow, W&B, DVC, Uvicorn, Microservices.

**AI-Assisted Development:** Claude Code, GitHub Copilot, Gemini.

## Projects

### Quantum Noise Mitigation with AI - Master's Thesis

2026

| Python, PyTorch Geometric, Qiskit Aer, Scikit-learn, W&B, DVC, GCP 🎓 MSc AI · UNIR

Pre-execution noise prediction for quantum circuits (20K circuits, real IBM calibration): Random Forest beat GNNs (GraphSAGE, GIN, GATv2) overall, with up to 81% error reduction; GNNs only won on larger, unseen circuits.

### RAG vs KAG - University Regulations Assistant

2026

| Python, txtai, networkx, Llama 3.3, Ollama

RAG vs KAG comparison on academic regulations: 3 retrieval engines (dense, BM25, hybrid) with txtai versus a Knowledge Graph (networkx) for multi-hop reasoning over implicit rules not present in the text, answering questions where plain RAG returned "I don't know".

### Hate Speech MLOps: Classification Pipeline & Model Serialization

2026

| Python, Scikit-learn, LightGBM, Optuna, SHAP, Joblib

Modular binary classification pipeline with Bayesian optimization (Optuna), explainability audit (SHAP) and production-ready *end-to-end* serialization. Reached 0.975 ROC-AUC and 0.91 F1-macro on test after detecting and removing a data leakage.

### Covertime: RF vs SVM Comparison at Scale

2026

| Python, Scikit-learn, Pandas, Matplotlib 🎓 MSc AI · UNIR

RF vs SVM comparison on UCI Covertime (581K records, 7 imbalanced classes): SVM (RBF) reached 0.76 F1-macro on a 50K subsample vs 0.68 for Random Forest on the full data. Feature selection with ANOVA and Chi<sup>2</sup>.

## Education

Universidad Internacional de La Rioja (UNIR) 🌐 [coursework]

Online

MSc in AI + Data Engineering Specialization

Nov 2025 – Present

**Master's Thesis:** Pre-execution noise prediction for *quantum circuits* with ML and graph neural networks.

University of Granada (UGR)

Granada, Spain

Bachelor's Degree in Mathematics

2021 – 2025

Completed within the standard timeframe (4 years) in the academic program **ranked No. 1 in Spain** (ShanghaiRanking GRAS 2025, Mathematics).

## Languages

Spanish / Armenian: Native | Valencian: C1 | English: B2